Probability and Statistics I

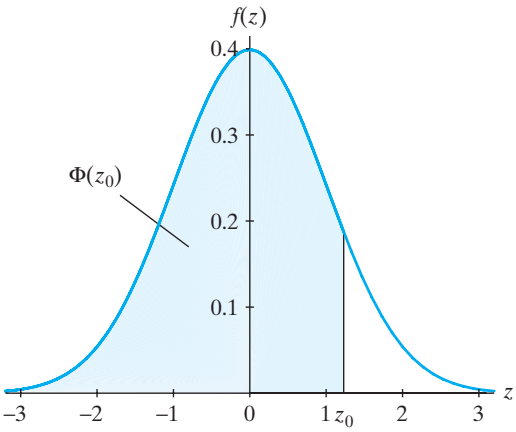
Mid-term Examination
SDS, CUHK(SZ)

June 28, 2025

Name: _____ Student ID: _____

Answer the multiple choice questions in the Answer Book. Please put all your answers on one page.

Table Va The Standard Normal Distribution Function

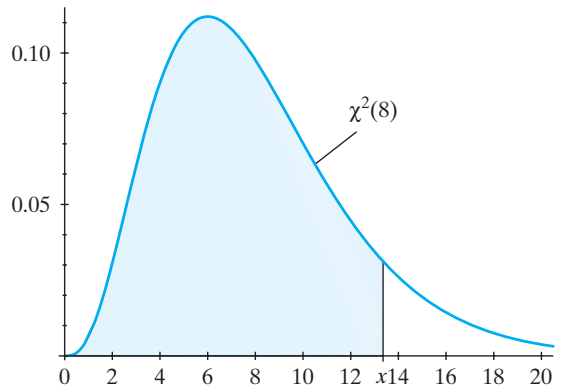


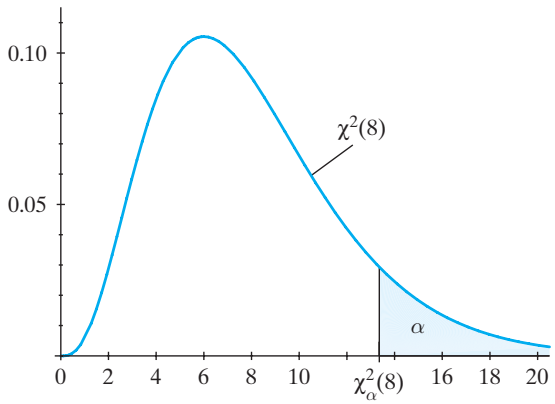
$$P(Z \le z) = \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-w^2/2}\, dw$$

$$\Phi(-z) = 1 - \Phi(z)$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
α	0.400	0.300	0.200	0.100	0.050	0.025	0.020	0.010	0.005	0.001
z_α	0.253	0.524	0.842	1.282	1.645	1.960	2.054	2.326	2.576	3.090
$z_{\alpha/2}$	0.842	1.036	1.282	1.645	1.960	2.240	2.326	2.576	2.807	3.291

Table IV The Chi-Square Distribution



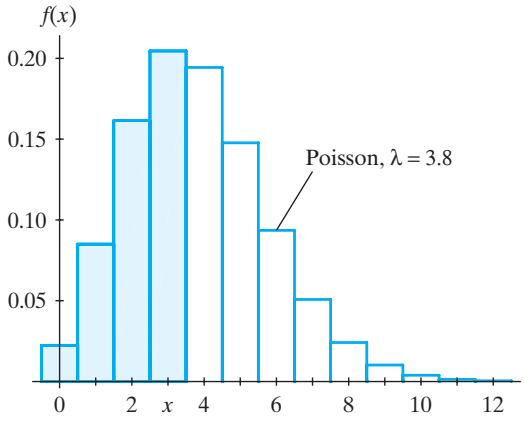


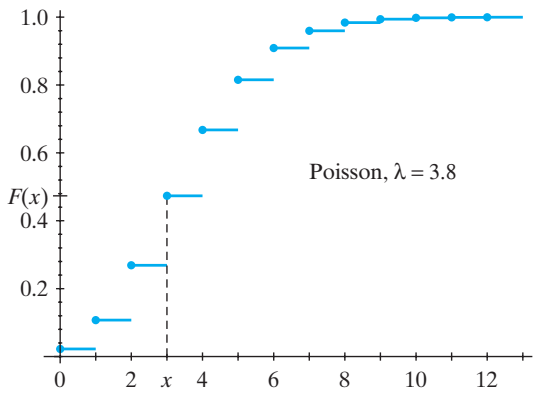
$$P(X \le x) = \int_0^x \frac{1}{\Gamma(r/2)2^{r/2}} w^{r/2-1} e^{-w/2} dw$$

	$P(X \le x)$							
	0.010	0.025	0.050	0.100	0.900	0.950	0.975	0.990
r	$\chi^2_{0.99}(r)$	$\chi^2_{0.975}(r)$	$\chi^2_{0.95}(r)$	$\chi^2_{0.90}(r)$	$\chi^2_{0.10}(r)$	$\chi^2_{0.05}(r)$	$\chi^2_{0.025}(r)$	$\chi^2_{0.01}(r)$
1	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.34
4	0.297	0.484	0.711	1.064	7.779	9.488	11.14	13.28
5	0.554	0.831	1.145	1.610	9.236	11.07	12.83	15.09
6	0.872	1.237	1.635	2.204	10.64	12.59	14.45	16.81
7	1.239	1.690	2.167	2.833	12.02	14.07	16.01	18.48
8	1.646	2.180	2.733	3.490	13.36	15.51	17.54	20.09
9	2.088	2.700	3.325	4.168	14.68	16.92	19.02	21.67
10	2.558	3.247	3.940	4.865	15.99	18.31	20.48	23.21
11	3.053	3.816	4.575	5.578	17.28	19.68	21.92	24.72
12	3.571	4.404	5.226	6.304	18.55	21.03	23.34	26.22
13	4.107	5.009	5.892	7.042	19.81	22.36	24.74	27.69
14	4.660	5.629	6.571	7.790	21.06	23.68	26.12	29.14
15	5.229	6.262	7.261	8.547	22.31	25.00	27.49	30.58
16	5.812	6.908	7.962	9.312	23.54	26.30	28.84	32.00
17	6.408	7.564	8.672	10.08	24.77	27.59	30.19	33.41
18	7.015	8.231	9.390	10.86	25.99	28.87	31.53	34.80
19	7.633	8.907	10.12	11.65	27.20	30.14	32.85	36.19
20	8.260	9.591	10.85	12.44	28.41	31.41	34.17	37.57
21	8.897	10.28	11.59	13.24	29.62	32.67	35.48	38.93
22	9.542	10.98	12.34	14.04	30.81	33.92	36.78	40.29
23	10.20	11.69	13.09	14.85	32.01	35.17	38.08	41.64
24	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98
25	11.52	13.12	14.61	16.47	34.38	37.65	40.65	44.31
26	12.20	13.84	15.38	17.29	35.56	38.88	41.92	45.64
27	12.88	14.57	16.15	18.11	36.74	40.11	43.19	46.96
28	13.56	15.31	16.93	18.94	37.92	41.34	44.46	48.28
29	14.26	16.05	17.71	19.77	39.09	42.56	45.72	49.59
30	14.95	16.79	18.49	20.60	40.26	43.77	46.98	50.89
40	22.16	24.43	26.51	29.05	51.80	55.76	59.34	63.69
50	29.71	32.36	34.76	37.69	63.17	67.50	71.42	76.15
60	37.48	40.48	43.19	46.46	74.40	79.08	83.30	88.38
70	45.44	48.76	51.74	55.33	85.53	90.53	95.02	100.4
80	53.34	57.15	60.39	64.28	96.58	101.9	106.6	112.3

This table is abridged and adapted from Table III in *Biometrika Tables for Statisticians*, edited by E.S.Pearson and H.O.Hartley.

Table III The Poisson Distribution





$$F(x) = P(X \le x) = \sum_{k=0}^{x} \frac{\lambda^k e^{-\lambda}}{k!}$$

	$\lambda = E(X)$									
x	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0	0.905	0.819	0.741	0.670	0.607	0.549	0.497	0.449	0.407	0.368
1	0.995	0.982	0.963	0.938	0.910	0.878	0.844	0.809	0.772	0.736
2	1.000	0.999	0.996	0.992	0.986	0.977	0.966	0.953	0.937	0.920
3	1.000	1.000	1.000	0.999	0.998	0.997	0.994	0.991	0.987	0.981
4	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.999	0.998	0.996
5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.999
6	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
x	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
0	0.333	0.301	0.273	0.247	0.223	0.202	0.183	0.165	0.150	0.135
1	0.699	0.663	0.627	0.592	0.558	0.525	0.493	0.463	0.434	0.406
2	0.900	0.879	0.857	0.833	0.809	0.783	0.757	0.731	0.704	0.677
3	0.974	0.966	0.957	0.946	0.934	0.921	0.907	0.891	0.875	0.857
4	0.995	0.992	0.989	0.986	0.981	0.976	0.970	0.964	0.956	0.947
5	0.999	0.998	0.998	0.997	0.996	0.994	0.992	0.990	0.987	0.983
6	1.000	1.000	1.000	0.999	0.999	0.999	0.998	0.997	0.997	0.995
7	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.999	0.999
8	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
x	2.2	2.4	2.6	2.8	3.0	3.2	3.4	3.6	3.8	4.0
0	0.111	0.091	0.074	0.061	0.050	0.041	0.033	0.027	0.022	0.018
1	0.355	0.308	0.267	0.231	0.199	0.171	0.147	0.126	0.107	0.092
2	0.623	0.570	0.518	0.469	0.423	0.380	0.340	0.303	0.269	0.238
3	0.819	0.779	0.736	0.692	0.647	0.603	0.558	0.515	0.473	0.433
4	0.928	0.904	0.877	0.848	0.815	0.781	0.744	0.706	0.668	0.629
5	0.975	0.964	0.951	0.935	0.916	0.895	0.871	0.844	0.816	0.785
6	0.993	0.988	0.983	0.976	0.966	0.955	0.942	0.927	0.909	0.889
7	0.998	0.997	0.995	0.992	0.988	0.983	0.977	0.969	0.960	0.949
8	1.000	0.999	0.999	0.998	0.996	0.994	0.992	0.988	0.984	0.979
9	1.000	1.000	1.000	0.999	0.999	0.998	0.997	0.996	0.994	0.992
10	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.999	0.998	0.997
11	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.999
12	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Multiple Choices (100 points)

- **4 points for each correct answer; -1.5 point for each incorrect answer; 0 points for no answer.**
- **For each question, only choose (at most) one out of four given choices (A,B,C and D). If you choose more than one choice in one question, your answer will be incorrect and 1.5 point will be deduced.**

1. If $P(A) = 0.5$, $P(B) = 0.6$, and $P(A \cap B) = 0.2$, find:

   (a) $P(A \cup B)$

   (b) $P(A' \cap B)$

   A. (a) 0.9, (b) 0.4
   B. (a) 0.8, (b) 0.3
   C. (a) 0.7, (b) 0.4
   D. (a) 0.8, (b) 0.2

   Solution: We are given:

   $$P(A) = 0.5, \quad P(B) = 0.6, \quad P(A \cap B) = 0.2$$

   (a) Find $P(A \cup B)$: We use the formula:

   $$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

   $$P(A \cup B) = 0.5 + 0.6 - 0.2 = 0.9$$

   (b) Find $P(A' \cap B)$: Use the identity:

   $$P(A' \cap B) = P(B) - P(A \cap B)$$

   $$P(A' \cap B) = 0.6 - 0.2 = 0.4$$

   The correct answer is $\boxed{A}$

2. A delivery drone starts from a main station S and travels through one of two regional hubs, A or B, before reaching one of three drop-off

locations: C, D, or E. The probabilities for the possible paths are:

$$\begin{aligned}
&P(\text{from } S \text{ to } A) = 0.7, \qquad P(\text{from } S \text{ to } B) = 0.3 \\
&P(\text{from } A \text{ to } C) = 0.3, \qquad P(\text{from } A \text{ to } D) = 0.3, \\
&P(\text{from } A \text{ to } E) = 0.4 \\
&P(\text{from } B \text{ to } C) = 0.5, \qquad P(\text{from } B \text{ to } D) = 0.2, \\
&P(\text{from } B \text{ to } E) = 0.3
\end{aligned}$$

Given that the drone arrived at location C, what is the conditional probability that it passed through hub A?

A. $\frac{5}{12}$ B. $\frac{7}{12}$ C. $\frac{21}{100}$ D. $\frac{3}{7}$

Solution: We are given:

$$P(\text{from } A) = 0.7, \quad P(\text{from } B) = 0.3$$

$$P(\text{arrived at } C|\text{from } A) = 0.3, \quad P(\text{arrived at C}|\text{from } B) = 0.5$$

Then,

$$\begin{aligned}
P(A \cap C) &= P(\text{arrived at } C|\text{from A})P(\text{from } A) = 0.21 \\
P(B \cap C) &= P(\text{arrived at } C|\text{from } B)P(\text{from } B) = 0.15
\end{aligned}$$

Thus,

$$P(C) = P(A \cap C) + P(B \cap C) = 0.36$$

Now calculate the conditional probability that the drone passed through A, given that it ended up at C:

$$P(A \mid C) = \frac{P(A \cap C)}{P(C)} = \frac{0.21}{0.36} = \frac{7}{12}$$

The correct answer is $\boxed{B}$

3. There are 4 fair coins and 2 unfair coins, all indistinguishable in appearance and feel.

- A fair coin lands heads with probability 0.5, and tails with 0.5.
- An unfair coin lands heads with probability 0.8, and tails with 0.2.

A coin is selected at random and flipped 4 times. The result is 2 heads and 2 tails.

What is the probability that the selected coin was fair?

A. $\frac{625}{753}$ B. $\frac{2}{3}$ C. $\frac{1}{2}$ D. $\frac{529}{625}$

Solution: Let:

- F: the event that the chosen coin is fair
- U: the event that the chosen coin is unfair
- D: the observed data is 2 heads in 4 flips

We want $P(F \mid D)$, the probability the coin is fair given 2 heads in 4 flips.

Step 1: Prior Probabilities

$$P(F) = \frac{2}{3}, \quad P(U) = \frac{1}{3}$$

Step 2: Likelihoods

Fair coin: $P(H) = 0.5$

$$P(D \mid F) = \binom{4}{2}(\frac{1}{2})^2(\frac{1}{2})^2 = \frac{3}{8}$$

Unfair coin: $P(H) = 0.8$

$$P(D \mid U) = \binom{4}{2}(\frac{8}{10})^2(\frac{2}{10})^2 = \frac{96}{625}$$

Step 3: Total Probability of Data

$$\begin{aligned} P(D) &= P(D \mid F) \cdot P(F) + P(D \mid U) \cdot P(U) \\ &= \frac{3}{8} \cdot \frac{2}{3} + \frac{96}{625} \cdot \frac{1}{3} = \frac{1}{4} + \frac{32}{625} = \frac{753}{2500} \end{aligned}$$

Step 4: Posterior Probability

$$P(F \mid D) = \frac{P(D \mid F) \cdot P(F)}{P(D)} = \frac{\frac{1}{4}}{\frac{753}{2500}} = \frac{625}{753}$$

The correct answer is $\boxed{A}$

4. You have 18 identical blue balls and 10 identical yellow balls. You place them randomly into 9 bins, each ball equally likely to go into any bin, and independently of the others. Compute the probability of the following event: Each bin contains at least one blue ball **AND** one yellow ball.

A. $\dfrac{\binom{17}{7} \cdot \binom{9}{8}}{\binom{27}{8} \cdot \binom{27}{8}}$ B. $\dfrac{\binom{17}{7} \cdot \binom{9}{7}}{\binom{26}{8} \cdot \binom{17}{8}}$

C. $\dfrac{\binom{16}{7} \cdot \binom{8}{7}}{\binom{27}{8} \cdot \binom{19}{7}}$ D. $\dfrac{\binom{17}{8} \cdot \binom{9}{8}}{\binom{26}{8} \cdot \binom{18}{8}}$

Solution:

- The total number of ways of placing blue balls is $\binom{18+9-1}{9-1} = \binom{26}{8}$.
- The total number of ways of placing yellow balls is $\binom{10+9-1}{9-1} = \binom{18}{8}$.
- The total number of ways of placing blue balls such that all bins must contain at least one blue ball is $\binom{(18-9)+9-1}{9-1} = \binom{17}{8}$.
- The total number of ways of placing yellow balls such that all

bins must contain at least one yellow ball is
$\binom{(10-9)+9-1}{9-1} = \binom{9}{8}$.

Thus, the probability is $\dfrac{\binom{17}{8}\cdot\binom{9}{8}}{\binom{26}{8}\cdot\binom{18}{8}}$.

The correct answer is $\boxed{D}$

5. Consider four events A, B, C, and D. Determine which of the following two statements is **CORRECT**, where T stands for true and F stands for false.

   (i) If A and B are independent, then

   $$P(A \cap B \cap C) = P(A)P(B)P(C|A \cap B)$$

   (ii) If A, B, C, and D are pairwise independent, then $A \cap B$ and $C \cap D$ are independent.

   A. (T, T)
   B. (T, F)
   C. (F, T)
   D. (F, F)

Solution: For the first statement,

$$P(A \cap B \cap C) = P(C|A \cap B)P(A \cap B) = P(C|A \cap B)P(A)P(B)$$

Thus, statement (i) is true.

For the second statement, we provide the following counterexample: Toss a fair coin 3 times. Let A be the event head on first, B be the event head on second, C the event head on third, and D the event all tosses are the same. Then we have pairwise independence. However, the probability of $A \cap B$ is $\frac{1}{4}$, the probability of $C \cap D$ is

$\frac{1}{8}$, and the probability of $(A \cap B) \cap (C \cap D)$ is $\frac{1}{8}$. Thus, they are not independent. Therefore, statement (ii) is false.

The correct answer is $\boxed{B}$.

6. A box contains 5 red, 4 blue, and 3 green balls. Two balls are drawn randomly without replacement.

   - (a) The probability that both are red is $\frac{2}{11}$ if the first ball drawn is red.
   - (b) The probability that the second ball is blue, given the first ball is green, is $\frac{4}{11}$.
   - (c) The probability of drawing two balls of different colors is $\frac{5}{33}$.
   - (d) The probability that at least one ball is green, given that one ball is red, is $\frac{1}{3}$.

   Find which statements are **CORRECT**:

   A. (a) and (b)
   B. (b) and (d)
   C. (c)
   D. (a), (c), and (d)

Solution: This question tests understanding of conditional probability.

(a) **Incorrect.** The probability that both are red, given the first is red, is:

$$P(\text{second is red} \mid \text{first is red}) = \frac{4}{11}.$$

(b) **Correct.** The probability that the second ball is blue, given the first is green:

$$P(\text{second is blue} \mid \text{first is green}) = \frac{4}{11}.$$

(c) **Incorrect.** The probability of drawing two balls of different colors is more complicated and certainly not $\frac{5}{33}$. It must be computed using combinations.

(d) **Correct.** The probability that at least one ball is green, given that one ball is red, is:

$$P(\text{at least one red}) = 1 - P(\text{no red}) = \frac{15}{22}$$

$$P(\text{one green and one red}) = \frac{5}{22}$$

$$P(\text{one green and one red} \mid \text{at least one red}) = \frac{1}{3}.$$

The correct answer: $\boxed{B}$

7. Suppose that we toss 2 fair dice. Define the following three events:

   a: The sum of the dice is 6.

   b: The first die equals 4.

   c: The sum of the dice is 7.

   Determine which of the following four statements is **CORRECT**:

   A. Events a and b are independent.
   B. Events b and c are independent.
   C. Events a, b, and c are mutually independent.
   D. Events a, b, and c are pairwise independent.

Solution: We compute:

$$\begin{aligned} P(a) &= \frac{5}{36}, \\ P(b) &= \frac{1}{6}, \\ P(c) &= \frac{6}{36} = \frac{1}{6}, \\ P(a \cap b) &= \frac{1}{36}, \quad \text{(Only the outcome (4,2))} \end{aligned}$$

$$P(b \cap c) = \frac{1}{36}, \quad \text{(Only the outcome (4,3))}$$

Now, test independence:

- $P(a \cap b) = \frac{1}{36} = P(a)P(b) = \frac{5}{36} \cdot \frac{1}{6} = \frac{5}{216} \neq \frac{1}{36} \Rightarrow a$ and b are not independent.
- $P(b \cap c) = \frac{1}{36} = P(b)P(c) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36} \Rightarrow b$ and c are independent.

Thus, only b and c are independent.

The correct answer is $\boxed{B}$.

8. Let A, B and C be three events. Which of the following statements is **FALSE**?

A. If the events A and B are mutually exclusive, then A and B are not necessarily independent.
B. If $A \subseteq B$, then A and B are NOT independent.
C. If $P(A) = 0.4$, $P(B) = 0.6$, and $P(A \cup B) = 0.76$, then A and B are independent.
D. Suppose A, B, C are pairwise independent. In addition, suppose A and $B \cup C$ are independent. Then A, B, C are mutually independent.

Solution: This problem tests understanding of independence between events.

(A) **True.** If A and B are mutually exclusive, then $P(A \cap B) = 0$. If $P(A) > 0$ and $P(B) > 0$, then $P(A \cap B) \neq P(A)P(B)$, so they are not independent.

(B) **False.** Consider the counterexample: let $A = \emptyset$, $B = S$. Then $P(A) = 0$, $P(B) = 1$, and $P(A \cap B) = 0 = P(A)P(B)$, so A and B are independent even though $A \subseteq B$. Hence, the statement is false.

(C) **True.** Use inclusion-exclusion:

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.4 + 0.6 - 0.76 = 0.24.$$

Since $P(A)P(B) = 0.4 \cdot 0.6 = 0.24$, A and B are independent.

(D) **True.** From the assumption that A, B, C are pairwise independent and A is also independent of $B \cup C$, we can use the identity:

$$P(A \cap B \cap C) = P(A)P(B)P(C),$$

proving mutual independence.

The correct answer is $\boxed{B}$

9. One fair six-sided dice (with faces numbered 1 to 6) is rolled. Let X be the random variable representing the number of a dice roll. The variance $\mathrm{Var}(X)$ is:

A. $\frac{49}{6}$ B. $\frac{35}{12}$ C. $\frac{49}{12}$ D. 7

Solution: Let X be the outcome of one dice. For a single die:

$$E[X] = \frac{1+2+3+4+5+6}{6} = \frac{7}{2}$$

$$E[X^2] = \frac{1^2+2^2+3^2+4^2+5^2+6^2}{6} = \frac{91}{6}$$

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$$

The correct answer is $\boxed{B}$

10. A factory has a 95% defect-free rate for its products. In a random sample of 20 products, let X be the number of defect-free products. The expectation $E(X)$ and variance $\mathrm{Var}(X)$ are:

A. $E(X) = 19$, $\mathrm{Var}(X) = 0.95$

B. $E(X) = 19$, $\text{Var}(X) = 0.475$
C. $E(X) = 19.5$, $\text{Var}(X) = 0.95$
D. $E(X) = 20$, $\text{Var}(X) = 0.475$

Solution: Since $X \sim \text{Binomial}(n = 20, p = 0.95)$,

$$E(X) = np = 20 \times 0.95 = 19$$

$$\text{Var}(X) = np(1-p) = 20 \times 0.95 \times 0.05 = 0.95$$

The correct answer is $\boxed{A}$

11. Consider a Bernoulli process with success probability p and define $q = 1 - p$. Let X be the number of Bernoulli trials until the r-th success and let N_k be a Binomial random variable with parameters $n = k$ and p.

(i) Is $P(X > k) = P(N_k \leq r - 1)$?

(ii) For $r = 1$ would this yield $P(X > k) = q^k$ for all $k \in \{1, 2, \ldots\}$?

(iii) Is it true that for $r = 1$, $P(\{X > k + l\} \mid \{X > k\}) = P(X > l)$ for all positive integers l?

(iv) For arbitrary r is it $P(X = k) = P(N_k \leq r-1) - P(N_{k-1} \leq r-1)$ correct for all $k \in \{r, r+1, \ldots, \}$?

A. (T, T, T, T)
B. (T, F, F, T)
C. (T, T, T, F)
D. (F, F, F, F)

Solution: (i) holds because the events $X > k$ and $N_k \leq r - 1$ are equivalent.

Part (ii) follows from (i) since $P(N_k \leq 0) = q^k$.

Part (iii) follows since the intersection of the events $\{X > k + l\}$ and $\{X > k\}$ is $\{X > k + l\}$, so $P(\{X > k + l\} \mid \{X > k\}) = P(\{X > k + l\})/P(\{X > k\}) = q^{k+l}/q^k = q^l = P(X > l)$.

Part (iv) is false since the correct answer is $P(X = k) = P(N_{k-1} \leq r - 1) - P(N_k \leq r - 1)$.

The correct answer is $\boxed{C}$

12. Let X_1, X_2, X_3 be independent Bernoulli random variables with success probability $p \in (0, 1)$.

(i) Is $P(X_1 + X_2 + X_3 = 3 \mid X_1 = 1) = p^2$?

(ii) Is $P(X_1 + X_2 + X_3 = 3 \mid X_1 + X_2 \geq 1) = \dfrac{p^2}{2 - p}$?

Determine which of the following statements is **CORRECT**, where T stands for true and F for false:

A. (F, F)
B. (T, F)
C. (T, T)
D. (F, T)

Solution: (i) Given $X_1 = 1$, we need $X_2 = 1$ and $X_3 = 1$ for the sum to be 3:

$$P(X_1 + X_2 + X_3 = 3 \mid X_1 = 1) = P(X_2 = 1 \cap X_3 = 1 \mid X_1 = 1)$$

By independence of X_1, X_2, X_3:

$$= P(X_2 = 1) \cdot P(X_3 = 1) = p \cdot p = p^2$$

Thus, the statement is **True**.

(ii) We need to compute:

$$\begin{aligned} &P(X_1 + X_2 + X_3 = 3 \mid X_1 + X_2 \geq 1) \\ =&\frac{P(X_1 + X_2 + X_3 = 3 \cap X_1 + X_2 \geq 1)}{P(X_1 + X_2 \geq 1)} \end{aligned}$$

Note that $X_1 + X_2 + X_3 = 3$ implies $X_1 = 1, X_2 = 1, X_3 = 1$, which automatically satisfies $X_1 + X_2 \geq 1$. Thus:

$$\text{Numerator} = P(X_1 = 1, X_2 = 1, X_3 = 1) = p^3$$

The denominator:

$$\begin{aligned}
P(X_1 + X_2 \geq 1) &= 1 - P(X_1 + X_2 = 0) \\
&= 1 - P(X_1 = 0)P(X_2 = 0) \\
&= 1 - (1-p)^2 \\
&= 1 - (1 - 2p + p^2) \\
&= 2p - p^2 \\
&= p(2-p)
\end{aligned}$$

Therefore:

$$P(X_1 + X_2 + X_3 = 3 \mid X_1 + X_2 \geq 1) = \frac{p^3}{p(2-p)} = \frac{p^2}{2-p}$$

Thus, the statement is **True**.

The correct answer is $\boxed{C}$

13. Let X follow a discrete uniform distribution on $\{a, \ldots, b\}$, where a and b are integers with $a \leq b$. The pmf of X is

$$P(X = x) = p(x) = \begin{cases} \frac{1}{b-a+1}, & \text{for } x \in \{a, \ldots, b\} \\ 0, & \text{otherwise} \end{cases}$$

Let $\mathbb{E}(X), \mathrm{Var}(X)$ and $M_X(t)$ denote the mean, variance, and moment generating function of X, respectively.

Find which one of the following statements is **CORRECT**.

A. (a) $\mathbb{E}(X) = \frac{a+b}{2}$, (b) $\mathrm{Var}(X) = \frac{(b-a+1)^2+1}{12}$, (c) $M_X(1) = \frac{e^a - e^b}{(b-a+1)(1-e)}$

B. (a) $\mathbb{E}(X) = \frac{b-a}{2}$, (b) $\mathrm{Var}(X) = \frac{(b-a+1)^2-1}{12}$, (c) $M_X(1) = \frac{e^a - e^{b+1}}{(b-a+1)(1-e)}$

C. (a) $\mathbb{E}(X) = \frac{a+b}{2}$, (b) $\mathrm{Var}(X) = \frac{(b-a+1)^2-1}{12}$, (c) $M_X(1) = \frac{e^a - e^{b+1}}{(b-a+1)(1-e)}$

D. (a) $\mathbb{E}(X) = \frac{b-a}{2}$, (b) $\mathrm{Var}(X) = \frac{(b-a+1)^2+1}{12}$, (c) $M_X(1) = \frac{e^a - e^b}{b-a+1}$

Solution:

(a) The expected value of X is

$$\mathbb{E}(X) = \sum_{x=a}^{b} \frac{x}{b-a+1} = \frac{1}{b-a+1} \sum_{x=a}^{b} x = \frac{a+b}{2}.$$

(b) Let $n = b-a+1$, define $Y = X-a+1$, then $Y \sim \text{Unif}\{1, 2, \ldots, n\}$. Since $\text{Var}(X) = \text{Var}(Y)$, we compute:

$$\mathbb{E}(Y) = \frac{n+1}{2}, \quad \mathbb{E}(Y^2) = \frac{(n+1)(2n+1)}{6}$$

$$\begin{aligned}
&\text{Var}(X) \\
=&\mathbb{E}(Y^2) - \mathbb{E}(Y)^2 \\
=&\frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 \\
=&\frac{n^2-1}{12}
\end{aligned}$$

Substituting $n = b - a + 1$, we get

$$\text{Var}(X) = \frac{(b-a+1)^2 - 1}{12}$$

(c) The moment generating function (MGF) of X is:

$$\begin{aligned}
&M_X(t) = \mathbb{E}(e^{tX}) \\
=&\sum_{x=a}^{b} \frac{e^{tx}}{b-a+1} = \frac{e^{ta}}{b-a+1}\left(1 + e^t + e^{2t} + \cdots + e^{(b-a)t}\right)
\end{aligned}$$

This is a geometric series, so we simplify:

$$\begin{aligned}
M_X(t) &= \begin{cases} \dfrac{e^{at}(1 - e^{(b-a+1)t})}{(b-a+1)(1-e^t)}, & t \neq 0 \\ 1, & t = 0 \end{cases} \\
&= \begin{cases} \dfrac{e^{at} - e^{(b+1)t}}{(b-a+1)(1-e^t)}, & t \neq 0 \\ 1, & t = 0 \end{cases}
\end{aligned}$$

The correct answer is $\boxed{(C)}$.

14. Suppose X follows a uniform distribution on the interval $[-2, 1]$, that is, $X \sim U[-2, 1]$. Let $Y = X^2$. Which one of the following is the cumulative distribution function of Y?

A. $F_Y(y) = \begin{cases} \dfrac{2\sqrt{y}}{3}, & 0 \le y \le 1, \\ \dfrac{1+\sqrt{y}}{3}, & 1 < y \le 4, \\ 0, & y < 0, \\ 1, & y > 4. \end{cases}$

B. $F_Y(y) = \begin{cases} \dfrac{1+\sqrt{y}}{3}, & 0 \le y \le 1, \\ \dfrac{\sqrt{y}}{2}, & 1 < y \le 4, \\ 0, & y < 0, \\ 1, & y > 4. \end{cases}$

C. $F_Y(y) = \begin{cases} \dfrac{1+\sqrt{y}}{3}, & 0 \le y \le 4, \\ 0, & y < 0, \\ 1, & y > 4. \end{cases}$

D. $F_Y(y) = \begin{cases} \dfrac{2\sqrt{y}}{3}, & 0 \le y \le 4, \\ 0, & y < 0, \\ 1, & y > 4. \end{cases}$

Solution:

- The cdf of X is

$$F_X(x) = \frac{x+2}{3}, \qquad -2 \le x \le 1.$$

- **Case** $0 \le y \le 1$ ($\Leftrightarrow -1 \le x \le 1$):

$$\begin{aligned} F_Y(y) &= P(Y \le y) = P(X^2 \le y) \\ &= P(-\sqrt{y} \le X \le \sqrt{y}) \\ &= F_X(\sqrt{y}) - F_X(-\sqrt{y}) \\ &= \frac{\sqrt{y}+2}{3} - \frac{-\sqrt{y}+2}{3} = \frac{2\sqrt{y}}{3}. \end{aligned}$$

- **Case** $1 < y \le 4$ ($\Leftrightarrow -2 \le x \le -1$):

$$\begin{aligned}
F_Y(y) &= P(Y \le y) \\
&= P(Y < 1) + P(1 \le Y \le y) \\
&= \frac{2}{3} + P(1 \le X^2 \le y) \\
&= \frac{2}{3} + P\big(-\sqrt{y} \le X \le -1\big) \\
&= \frac{2}{3} + \big[F_X(-1) - F_X(-\sqrt{y})\big] \\
&= \frac{2}{3} + \left(\frac{-1+2}{3} - \frac{-\sqrt{y}+2}{3}\right) \\
&= \frac{1+\sqrt{y}}{3}.
\end{aligned}$$

- **Combine the two pieces** (together with the obvious tails):

$$F_Y(y) = \begin{cases} \dfrac{2\sqrt{y}}{3}, & 0 \le y \le 1, \\ \dfrac{1+\sqrt{y}}{3}, & 1 < y \le 4, \\ 0, & y < 0, \\ 1, & y > 4. \end{cases}$$

The correct answer is $\boxed{(A)}$.

15. A random variable X with parameter $\theta \in \mathbb{R}$ is said to belong to the *exponential family* if its probability mass (or density) function can be written as

$$f(x) = h(x)\, e^{\theta x - A(\theta)},$$

where $h(x)$ is free of θ and $A(\theta)$ is free of x. Let

$$M_X(t) = \mathbb{E}\big[e^{tX}\big]$$

be the moment–generating function of X. Exactly one of the following formulas is **CORRECT**. Which one is it?

A. $M_X(t) = e^{A(\theta) - A(\theta - t)}$

B. $M_X(t) = e^{A(\theta+t)-A(\theta)}$
C. $M_X(t) = e^{A(\theta)+A(t)}$
D. $M_X(t) = e^{t\,A(\theta)}$

Solution: *Derivation:*

$$\begin{aligned} E\big(e^{tX}\big) &= \int e^{tx}\, h(x)\, e^{\theta x - A(\theta)}\, dx \\ &= e^{-A(\theta)} \int h(x)\, e^{(\theta+t)x}\, dx \\ &= e^{A(\theta+t)-A(\theta)} \int h(x)\, e^{(\theta+t)x - A(\theta+t)}\, dx \\ &= e^{A(\theta+t)-A(\theta)}, \end{aligned}$$

where the last integral equals 1 because it is the pdf of the same exponential-family form with parameter $\theta + t$.

Hence the only formula that matches this result is option $\boxed{(B)}$.

16. The number of times that an individual contracts a cold in a given year is a **Poisson random variable** with mean $\theta = 6$. Suppose a new wonder drug reduces the Poisson mean to $\theta = 4$ for 60% of the population; for the other 40% of the population the drug has no effect on colds. If an individual takes the drug for a year and records 3 colds in that time, how likely is it that the drug is beneficial for him/her?

A. 0.82
B. 0.77
C. 0.56
D. 0.69

Solution:

Let X be the number of colds contracted in a year. Define event $E =$ "drug is effective", so

$$P(E) = 0.6, P(E^c) = 0.4.$$

$$X \mid E \sim \text{Poisson}(4), X \mid E^c \sim \text{Poisson}(6).$$

$$P(X = 3 \mid E) = \frac{e^{-4}4^3}{3!} = \frac{32e^{-4}}{3},$$

$$P(X = 3 \mid E^c) = \frac{e^{-6}6^3}{3!} = 36e^{-6}.$$

Using the law of total probability,

$$\begin{aligned} P(X = 3) &= P(X = 3 \mid E)\, P(E) + P(X = 3 \mid E^c)\, P(E^c) \\ &= 0.6\left(\frac{32e^{-4}}{3}\right) + 0.4\left(36e^{-6}\right) \\ &= 6.4e^{-4} + 14.4e^{-6}. \end{aligned}$$

Applying Bayes' theorem,

$$\begin{aligned} P(E \mid X = 3) &= \frac{P(X = 3 \mid E)\, P(E)}{P(X = 3)} \\ &= \frac{6.4e^{-4}}{6.4e^{-4} + 14.4e^{-6}} \approx 0.7666 \approx 0.77. \end{aligned}$$

Hence, the correct choice is $\boxed{\text{B}}$.

17. Consider the nonnegative random variable X with the pdf

$$f_X(x) = xe^{-x^2} + \alpha e^{-\beta x}$$

Here, α and β are constants to be determined.

Find α and β such that the 80th percentile is at the point $\pi_{0.8} = 1$.

A. $\alpha = 1, \beta = 2$
B. $\alpha = -\log(0.4 - e^{-1})/2, \beta = 2\alpha$
C. $\alpha = 1 - 0.4e, \beta = 1 - 0.2e$
D. $\alpha = 0.2e, \beta = 0.4e$

Solution: Normalization:

$$1 = \int_0^\infty \left(xe^{-x^2} + \alpha e^{-\beta x}\right) dx = \frac{1}{2} + \alpha/\beta$$

80th percentile implies

$$\begin{aligned} 0.8 &= \int_0^1 \left(x e^{-x^2} + \alpha e^{-\beta x} \right) dx \\ &= \frac{1}{2}(1 - e^{-1}) + \alpha(1 - e^{-\beta})/\beta \\ &= \frac{1}{2}\left(2 - e^{-1} - e^{-\beta}\right). \end{aligned}$$

Solving the two equations gives $\beta = 2\alpha = -\log(0.4 - e^{-1}) > 0$. Hence, the correct choice is $\boxed{\text{B}}$.

18. Suppose that X is a continuous random variable with pdf

$$f(x) = \begin{cases} x, & 0 \le x < 1 \\ 2 - x, & 1 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$$

Define a random variable $Y = 4X^2 + 3$. What is the value of $P(2 < Y \le 12)$?
A. 7/8.
B. 0.
C. 3/4.
D. 3/2.

Solution: The event $2 < Y \le 12$ reduces to $X \in [-3/2, 3/2]$ and thus to $X \in [0, 3/2]$. Hence

$$P(2 < Y \le 12) = \int_1^{3/2} (2 - x)dx + \int_0^1 x dx = 7/8.$$

Hence, the correct choice is $\boxed{\text{A}}$.

19. Consider the following 4 statements:

(i) If X follows a Gamma distribution with parameter $\alpha = 2$ and $\theta = 1$, then $E[X^4] = 120$.

(ii) Consider the Gamma function with a positive integer α, then we have $\Gamma(\alpha) = (\alpha - 1)!$.

(iii) If $X \sim N(3, 1)$, then $E[X^3] = 27$.

(iv) If X follows a Chi-square distribution with 1 degree of freedom, then $P(X \geq 2.706) = 0.9$.

A. All statements are true.
B. Only (i), (ii) and (iii) are true.
C. Only (iii) and (iv) are true.
D. Only (i) and (ii) are true.

Solution:

(i). Using the moment generating function of gamma distribution, we have $E[X^4] = \alpha(\alpha+1)(\alpha+2)(\alpha+3)\theta^4 = 120$.
(ii). This is a property of Gamma function and it can be proved by using integration by part.
(iii). Let $X = Z + 3$, where $Z \sim N(0, 1)$. Notice that $E[Z] = 0$, $E[Z^2] = 1$ and $E[Z^3] = 0$, we have $E[X^3] = E[Z^3 + 9Z^2 + 27Z + 27] = 36$.
(iv). This can be checked through the table. The correct one is $P(X \leq 2.706) = 0.9$.
Therefore, only (i) and (ii) are true.

Hence, the correct choice is $\boxed{\text{D}}$.

20. Consider the moment generating function of X, $M_X(t) = e^{3t+9t^2}$. And Y has the moment generating function, for $t < 3$, $M_Y(t) = \frac{3e^{2t}}{3-t}$.
Statement 1 : Y has $E[Y]$ of 79.
Statement 2 : Y has $Var(Y)$ of 13.
Statement 3 : X follows normal distribution $N(3, 9)$.
How many statements are **CORRECT**?
A. 0.
B. 1.
C. 2.
D. 3.

Solution:

$E[Y] = \frac{7}{3}$, and $Var(Y) = \frac{1}{9}$. The moment generating function of a normal distribution is $M(t) = e^{\mu t + \sigma^2 t^2/2}$, so that $X \sim N(3, 18)$.

Hence, the correct choice is $\boxed{\text{A}}$.

21. Suppose that $X \sim N(3, 4)$, find the smallest d below such that $P(X < d) \geq 0.8$.
A. 5.56.
B. 4.70.
C. 3.98.
D. 1.31.

Solution:

$$\begin{aligned} P(X < d) =& P(\frac{X-3}{2} \leq \frac{d-3}{2}) \\ =& \Phi(\frac{d-3}{2}) \geq 0.8. \end{aligned}$$

Check the table we know that $\Phi(0.84) = 0.8$. Thus $\frac{d-3}{2} \geq 0.84$, $d \geq 4.68$.

Hence, the correct choice is $\boxed{\text{B}}$.

22. Suppose that a restaurant everyday (24 hours) receives 720 complaints on average. It is assumed that the number of complaints received follow an approximate Poisson process. Suppose that X (in month) is the time it takes for the restaurant to reply to a complaint, and X follows a gamma distribution with mean $\frac{3}{2}$ months and standard deviation $\sqrt{\frac{3}{4}}$ months.

Statement 1 : The probability that the restaurant will have to wait longer than 21.06 minutes for the 7-th complaint is 0.05.
Statement 2 : Let $Y = 3X$. Then Y has a gamma distribution with the moment generating function $(\frac{1}{1-\frac{3}{2}t})^3$.

Which statement is **TRUE**?
A. Statements 1 and 2.
B. Statements 1 only.
C. Statements 2 only.
D. None of them.

Solution: 1. The mean rate of complains per minute is $\lambda = 1/2$. Thus $\theta = 2$ and $\alpha = r/2 = 7$. If Z denotes the waiting time until the 7th complaint, then Z is $\chi^2(14)$. $P(Z > 21.06) = 0.1$.
2. $M_X(t) = (\frac{1}{1-\frac{1}{2}t})^3$, $M_Y(t) = E[e^{tY}] = M_X(3t) = (\frac{1}{1-\frac{3}{2}t})^3$.

Hence, the correct choice is $\boxed{\text{C}}$.

23. Suppose that $f_1(x)$ is the pdf of the standard normal distribution, and $f_2(x)$ is the pdf of the uniform distribution over $[0, 4]$. Let $f(x)$ be a function defined as

$$f(x) = \begin{cases} af_1(x), & x \leq 0 \\ bf_2(x), & x > 0 \end{cases}$$

where $a > 0$ and $b > 0$. If $f(x)$ is a pdf, then which one of the following equalities must hold?
A. $3a + 4b = 6$.
B. $3a + 2b = 4$.
C. $a + 2b = 2$.
D. $a + b = 2$.

Solution: According to definition of probability density function: $\int_{-\infty}^{\infty} f(x)dx = 1$. Since $f_1(x) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{x^2}{2})$, and

$$f_2(x) = \begin{cases} \frac{1}{4}, & 0 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}.$$

We have

$$\begin{aligned} \int_{-\infty}^{\infty} f(x)dx &= \int_{-\infty}^{0} af_1(x)dx + \int_{0}^{\infty} bf_2(x)dx \\ &= a\frac{1}{2} + b = 1. \end{aligned}$$

Hence, the correct choice is $\boxed{\text{C}}$.

24. Suppose that the moment-generating function $M_X(t)$ and $M_Y(t)$ of the continuous random variable X and Y has the property $M_X(t) = e^t M_Y(-t)$ for all t.

    Which of the following statement is **CORRECT**?
    A. $X + Y = 1$.
    B. X must follow a uniform distribution on $[0, 1]$.
    C. $Var(X) = Var(Y)$.
    D. $E[X] = 1/2$.

Solution: $E(e^{tX}) = e^t E(e^{-tY}) = E(e^{t(1-Y)})$ and thus $M_X(t) = M_{1-Y}(t)$ for all t. Since the moment generating function determines uniquely the probability distribution, it follows that the random variable X has the same distribution as the random variable $1 - Y$. Therefore $Var(X) = Var(1 - Y) = Var(Y)$. The other statements are not necessarily true.

Hence, the correct choice is $\boxed{\text{C}}$.

25. Let X be the lifetime (measured in hours) of electronic device A and Y be the lifetime of device B. Suppose X follows an exponential distribution with rate 0.1 and Y follows a uniform distribution on $[0, 30]$. We have the following two statements:

    (i) Device A has a higher probability of having at least 10 hours of lifetime compared to device B.

    (ii) If both devices have functioned for 15 hours, then device A has a higher probability to have at least 10 hours of remaining lifetime, compared to device B.

    Find which one of the following statements is **CORRECT**?

    A. Both statements are correct.
    B. Both statements are incorrect.

C. Statement (i) is correct and (ii) is not.
D. Statement (ii) is correct and (i) is not.

Solution: It is clear that $P(X > 10) = e^{-1}$ and $P(Y > 10) = 2/3 > P(X > 10)$, indicating that device B has a larger probability of having lifetime no smaller than 10. Therefore statement (i) is not correct. Condition on the event that $X > 15$, the probability that X has at least 10 hours of remaining lifetime is $P(X > 25|X > 15) = e^{-1}$ and similarly, $P(Y > 25|Y > 15) = 1/3 < e^{-1}$. Therefore, statement (ii) is correct.

Hence, the correct choice is $\boxed{\text{D}}$.